

Who Prices Your Product and Why Does It Matter?

The Joint Effects of Pricing Agent and Personalization Basis on
Consumer Reactions regarding Perceived Pricing Fairness

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Abstract

Due to the increasing availability of consumer data, firms increasingly use algorithmic pricing agents for personalised pricing. Prior research has examined consumer responses to different pricing agents and types of consumer data, particularly in relation to perceived pricing fairness. However, limited research has examined their joint influence. This study therefore investigates to what extent the AI fairness advantage in personalised pricing depends on whether prices are based on behavioural rather than demographic consumer data.

I conducted a 2×2 between-subjects vignette experiment in the travel insurance context. Participants evaluated personalised prices determined either by an AI system or a human pricing analyst, using either behavioural or demographic information. The study included 65 participants, with perceived pricing fairness as the primary outcome and purchase intention as a secondary outcome.

The primary analyses did not provide statistically significant support for the hypothesised effects of pricing agent, personalisation basis, or their interaction on perceived pricing fairness. Although the estimated differences were in the predicted direction, they were not statistically significant. Behavioural personalisation was, however, associated with significantly higher purchase intention than demographic personalisation. A supplementary robustness analysis among participants who correctly identified the pricing agent showed a significant AI–human difference in perceived fairness in the predicted direction, but this finding does not replace the primary analysis.

Overall, the findings provide inconclusive evidence for the proposed relationships and highlight the need for further research, using larger samples and different personalised pricing contexts.

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1. Introduction

Imagine booking a travel insurance online for a holiday you have been planning for years. After answering mandatory questions about your trip, an insurance quote of €120 is established. After discovering that personal data was used to determine the price, you start wondering whether the applied pricing procedure is fair.

Would your verdict be different if the price had been determined by an AI system rather than a human pricing agent? Furthermore, would your perception of fairness change depending on whether the insurer used demographic information (age, gender, or location) or behavioural information (previous travel experiences) to determine the price?

Currently, companies in sectors such as aviation and insurance increasingly use both human and algorithmic pricing mechanisms to determine prices of goods and services. Although human pricing agents remain relevant, a shift can be seen as organizations increasingly replace the pricing workforce with AI agents (Duani et al., 2024). This phenomenon is taking place due to companies having access to a higher degree of digitally obtained data than ever before. While it is favourable for companies to use personalised pricing in order to maximize the organizational utility (charging closer to maximum willingness to pay consumers), the technological pricing revolution also creates some concerns. Consumers don't respond the same to this matter as the views and expectations of individuals differ on aspects including pricing fairness, transparency of the pricing process and (possible) price discrimination (Duani et al., 2024; Zhang and Cheng, 2024).

Before going into the effects of how personalized pricing procedures are applied, consumers consider a fair process leading to a pricing decision extremely relevant (Xia et al., 2004). The perceived pricing fairness of the process affects the degree to which consumers accept personalized pricing (Hufnagel et al., 2022). As part of the fairness process, the study of Campbell (1999) demonstrates that consumers evaluate the pricing process and establish an opinion on whether firms act out of self-interest. As the selection of a human or algorithmic pricing agent may influence the consumer fairness perception (in relationship to the pricing procedure), it is therefore relevant for this research.

This master thesis expands on the current knowledge that individuals evaluate pricing procedures differently when these are established by either a human pricing agent or an algorithmic pricing model (Duani et al., 2024; Zhang and Cheng, 2024). When pricing is

based on demographic information of the target group, consumers perceive algorithmic pricing as fairer and therefore more justified. These results can be explained as consumers feel less judged and have the belief that firms act less exploitative (Duani et al., 2024). The study of Zhang and Cheng (2024) elaborated further that different responses to prices set by humans or algorithms also depends on the pricing context.

Other research also shows that the type of consumer data used influences perceived pricing fairness, not even taking the end price into consideration (Priester et al., 2020; Heidary et al., 2024). Generally speaking, the usage of certain demographic data aspects for pricing purposes (including information regarding age & location) are viewed negatively by consumers and therefore creates price discrimination. These elements are uncontrollable and therefore don't feel justified (Priester et al., 2020; Heidary et al., 2024).

Therefore, this study aims to integrate both research directions (the AI versus the human agent and datatype personalized pricing selection) when considering perceived pricing fairness perceptions. Existing studies evaluated the effects of different pricing agents and datatype on perceived pricing fairness results independently. To the best of our knowledge, limited research has examined the joint effect of both pricing agent and datatype when measuring perceived pricing fairness outcomes. The study of Duani et al. (2024) also demonstrated the need for extra research on what is the effect of using behavioural elements (browsing history or social media activity) on consumer price discrimination standards.

Based on the research gap, I formulate the following research question:

To what extent does the AI fairness advantage in personalized pricing depend on whether prices are based on behavioral rather than demographic consumer data?

This master thesis contributes to the personalised pricing literature by integrating research on pricing agents and consumer data types within a single experimental framework. Moreover, it examines whether the relationship between the pricing agent and perceived pricing fairness depends on the type of consumer data used for personalised pricing. This study therefore provides an empirical examination of relationships that previous research has largely investigated separately.

The remainder of this paper is organised as follows. Section 2 reviews relevant literature while Section 3 develops the theory and hypotheses. Section 4 presents the experiment,

including the applied research method, empirical results and a study-specific discussion. Lastly, Section 5 contains the general discussion of this study

The anonymized data, data dictionary, survey materials, analysis code, statistical outputs, and additional documentation associated with this study are available through the project's Open Science Framework (OSF) repository:

https://osf.io/g2fb8/overview?view_only=0a8f16700afb42ef8ab61a1dc14e224a

2. Relevant Literature

This section reviews relevant literature on price fairness and attribution, AI versus human pricing decisions, and personalised pricing based on different types of consumer data.

2.1 Price Fairness and Attribution Theory

Price fairness can be defined as “judgments involving a comparison of a price or procedure with a pertinent standard, reference, or norm” (Xia et al., 2004). In other words, consumers need to compare prices in comparison with a set reference point to establish whether the previously paid price can be considered fair. These reference points include comparing prices against what other consumers paid. Furthermore, the study refers to this as distributive justice elements and price evaluations are therefore comparative. Ultimately, a price can be perceived as reasonable, acceptable or justifiable (Xia et al., 2004).

Regardless of the price, the previously mentioned study also contributes that besides the price of a product or service, procedural justice of how the price is established is relevant as well when considering the consumer evaluation of perceived pricing fairness. This distinction between distributive and procedural fairness is relevant for personalised pricing, where consumers often judge the fairness of the entire pricing process rather than the price outcome alone.

The importance of pricing procedural justice is aligned with the dual entitlement principle, introduced in the paper of Kahneman et al. (1986). In the opinion of consumers, both buyers and sellers have the right to use a reference point and to move the willingness to pay or pricing decisions based on referring to events in the industry. The dual entitlement theory also suggests that whenever a firm is moving prices just for increasing profits without having any other valid reason (price competitors changing or shortage in the market), it violates the community standard of fairness regarding the rights of consumers.

Building on the dual entitlement theory, Bolton et al. (2003) shows that consumers evaluate pricing decisions by using reference points such as previous prices, competitor prices and production costs. Together, these studies suggest that consumers judge pricing fairness by considering whether the pricing procedure appears justified rather than evaluating just the price outcome alone.

The study of Xia et al. (2004) also discusses the importance of the attribution theory in connection with pricing, as consumers raise an opinion on pricing decisions while not understanding the entire pricing process design, which in simple terms can be described as consumers using incomplete information. This theoretical framework can also be connected to perceived pricing fairness.

The attribution theory is defined as the process of how consumers obtain information about a product failure while afterwards giving a behavioural response to the firm based on the cause attached. It is important to note that responses of consumers can be subjective as individuals do not have access to all necessary information for supporting a certain belief (Folkes, 1984).

Attribution theory explains how consumers infer the causes of organisational actions when complete information is not available (Folkes, 1984). Rather than evaluating only the pricing outcome, consumers seek to understand why a particular pricing decision was made by attributing it to several underlying causes. These causal attributions influence how consumers evaluate the fairness of the pricing procedure and shape their subsequent behavioural responses.

Research applying attribution theory to pricing shows that fairness evaluations of consumers depend partly on the causes and motives they attribute to pricing decisions. Vaidyanathan and Aggarwal (2003) found that price increases were perceived as less fair when attributed to controllable organisational decisions (internal locus) rather than unavoidable external circumstances. Similarly, Campbell (1999) showed that consumers respond negatively when price increases are attributed to self-interested profit maximisation rather than legitimate business motives. Together, these studies suggest that perceived pricing fairness depends on the motives consumers attribute to pricing decisions rather than on the price outcome alone.

Overall, attribution theory provides a useful framework for understanding how consumers evaluate pricing decisions made by firms. Consumer judgments are shaped by the causes, responsibilities, and motives they attribute to pricing decisions, which subsequently influence perceptions of pricing fairness. This theoretical perspective provides the foundation for

examining whether fairness perceptions differ when prices are determined by AI pricing agents rather than human pricing agents.

2.2 AI versus Human Pricing Decisions

Prior research on algorithmic decision-making suggests that consumer responses to AI are context-dependent. Consumers may prefer algorithmic advice in some decision contexts, particularly when algorithms are perceived as more capable, but may be less willing to rely on AI when they believe human judgement is better suited to individual circumstances (Logg et al., 2019; Longoni et al., 2019). This context dependence is particularly relevant to personalised pricing, where consumers evaluate not only the outcome but also how the pricing decision is made.

Within personalised pricing, existing evidence points towards an AI fairness advantage. Duani et al. (2024) find that consumers perceive prices determined by AI as fairer than those determined by human pricing analysts when demographic consumer data are used. The authors argue that this effect arises because algorithms are perceived as less judgemental and less driven by self-interested motives than human decision makers. This explanation is consistent with Campbell's (1999) findings that consumers evaluate pricing decisions by inferring the firm's underlying motives. Because AI is perceived as less likely to pursue opportunistic profit maximisation, algorithmic prices appear more objective and justified, translating into a higher perception of pricing fairness and stronger purchase intentions (Duani et al., 2024).

Nevertheless, the AI fairness advantage is not unconditional. Zhang and Cheng (2024) demonstrate that consumers' fairness perceptions depend on the pricing context. When consumers receive disadvantageous prices, AI-generated prices are perceived as fairer because algorithms are viewed as lacking intentional self-interest. In contrast, when consumers receive advantageous prices, this advantage disappears because consumers attribute favourable outcomes to personal luck rather than to the pricing agent. These findings suggest that consumers' attributions regarding intentionality and firm motives determine whether an AI fairness advantage emerges.

Existing research suggests that AI pricing is generally perceived as fairer than human pricing because consumers attribute fewer self-interested motives to algorithmic decision makers. These findings also indicate that the AI fairness advantage may depend on contextual factors. While previous studies have examined contextual factors such as pricing outcomes, limited attention has been given to whether the type of consumer data used to personalise

prices similarly shapes fairness perceptions. This potential boundary condition is examined in the following section.

2.3 Personalized Pricing and Data Type

Personalised pricing refers to the practice of adjusting prices for individual consumers or consumer segments based on personal or situational information (Hufnagel et al., 2022; Priester et al., 2020). Unlike dynamic pricing, where prices fluctuate for all consumers in response to market conditions, personalised pricing tailors prices according to consumer-specific information such as purchase history, location or other personal characteristics. Enabled by advances in information technology, this practice allows firms to better match prices to the willingness to pay of consumers, thereby increasing profits.

Despite its economic benefits for firms, personalized pricing is generally perceived negatively by consumers. Research consistently shows that personalized pricing reduces perceived pricing fairness, thereby weakening the relationship between consumers and firms (Ohlwein and Bruno, 2025). Supporting this finding, Hufnagel et al. (2022) demonstrate that consumers respond negatively to both disadvantageous and advantageous personalized pricing outcomes. Lower perceptions of pricing fairness also translate into adverse behavioural responses, including increased negative word-of-mouth, more switching intentions, and lower purchase intentions (Hufnagel et al., 2022).

However, consumers do not respond equally to all forms of consumer data used in personalised pricing. Priester et al. (2020) show that consumers perceive pricing based on broader consumer segments as fairer than fully individualised pricing because less personally sensitive information is used. Similarly, consumers evaluate purchase-based data more favourably than location-based data, as purchase history is considered a more legitimate basis for price differentiation (Priester et al., 2020).

Supporting these findings, Heidary and Pluut (2025) demonstrate that consumers perceive behavioural data as fairer than demographic data because behavioural information is viewed as more relevant and expected within commercial interactions. This distinction is particularly relevant to personalised pricing because the information used to determine an individual price forms part of the pricing procedure that consumers evaluate. Their findings further show that procedural fairness has a stronger influence on the behavioural and emotional responses from consumers than distributive fairness. Therefore, consumers evaluate not only whether the resulting price is favourable but also whether the procedure used to establish that price is perceived as fair. Even when consumers receive a favourable price,

concerns about an unfair pricing procedure can still result in negative behavioural and emotional responses. Consequently, the type of consumer information used in personalised pricing may be particularly important because it can influence how consumers evaluate the fairness of the pricing procedure.

Heidary et al. (2024) further demonstrate that consumers evaluate different types of consumer data differently when used for personalised pricing. Their findings indicate that prices based on demographic information, such as ethnicity, sexual orientation and intelligence, are perceived as less fair than prices based on behavioural information, including browsing behaviour, device type and battery level. These findings are consistent with Shaddy et al. (2026), who similarly find that behavioural targeting is perceived more favourably than demographic targeting in a broader marketing context. Together, these results suggest that the perceived legitimacy of the data used to personalise prices plays an important role in shaping perceived pricing fairness.

Lastly, the attribution theory provides a useful explanation for these differences in fairness & trust perceptions. Sun and Tang (2021) argue that consumers evaluate data-driven pricing by attributing responsibility (locus of causality) and controllability to firms. When consumers believe firms intentionally exploit personal data for profit or perceive that they have little control over how their information is used, fairness perceptions and consumer trust decline. On the contrary, greater perceived controllability reduces negative fairness perceptions. These findings suggest that consumers evaluate personalised pricing not only according to the data being used but also according to the perceived legitimacy of that data use.

In conclusion, existing research demonstrates that consumers evaluate personalised pricing differently depending on the type of consumer data used in combination with the perceived legitimacy of the pricing procedure. While previous studies have compared different categories of consumer data, limited research has examined whether the fairness advantage of AI over human pricing analysts depends on whether personalised prices are based on behavioural vs demographic data. Addressing this gap forms the central objective of the present study and shapes the hypothesis developed in the next section.

3. Theory

Based on the literature reviewed previously, this section develops the hypotheses regarding the effects of pricing agent and personalization basis on perceived pricing fairness. First, the independent effects of pricing agent and personalization basis are discussed, followed by their potential interaction.

3.1 AI versus human pricing decisions

The first hypothesis builds on previous findings of an AI fairness advantage in personalised pricing. Examining whether this effect is also observed in the present study provides the foundation for assessing whether the strength of this advantage depends on the type of consumer data used.

Consumers evaluate not only the final price but also the procedure through which the price is determined (Campbell, 1999; Xia et al., 2004). The pricing agent is an important component of this procedure as consumers may attribute different motives and characteristics to human and AI decision makers. Human pricing agents may be perceived as more likely to act in their own self-interest or judge consumers, whereas AI systems may be perceived as more consistent, factual, rational, and less intentionally exploitative (Duani et al., 2024; Zhang and Cheng, 2024). These perceptions can influence how consumers interpret the legitimacy and fairness of the resulting personalised price.

Taken together, these findings suggest that consumers may perceive AI-generated personalised pricing as fairer than pricing determined by a human pricing analyst, because AI is less likely to be associated with self-interested or judgemental motives.

H1: AI-generated personalized pricing procedures are perceived as fairer than human-generated personalized prices.

3.2 Personalized Pricing and Data type

Besides the pricing agent, consumers may also evaluate the type of consumer information used to personalise a price. The basis on which a price is determined can influence perceptions of fairness, because consumers may consider whether the information used by the firm is relevant and acceptable for differentiating prices between consumers (Heidary et al., 2024; Priester et al., 2020).

A key distinction concerns the controllability of the information used for price personalisation. When prices are based on characteristics that consumers cannot easily control or change, consumers may perceive the resulting price differences as less fair (Sun and Tang, 2021). Demographic characteristics such as gender, age, or location can therefore be perceived as less acceptable bases for price differentiation, particularly when consumers do not consider these characteristics relevant to the pricing decision.

In contrast, behavioural information, such as browsing history or previous purchase behaviour, is more closely connected to consumers' own actions and may therefore be perceived as more relevant to the pricing decision. Priester et al. (2020), for example, found that location-based personalisation was perceived as less fair than personalisation based on purchase behaviour. More broadly, previous research comparing different types of consumer information has found that behavioural information is generally evaluated more favourably than demographic information when used for personalised pricing (Heidary et al., 2024; Shaddy et al., 2026; Sun and Tang, 2021). Consumers may perceive behavioural information as a more legitimate basis for price differentiation because their behaviour is at least partly controllable and can be changed through their own actions.

Therefore, consumers are expected to perceive personalised pricing based on behavioural information as fairer than personalised pricing based on demographic information.

H2: Personalized pricing procedures based on behavioral information are perceived as fairer than prices based on demographic information.

3.3 The Moderating Role of Personalization Basis

The preceding hypotheses suggest that both the pricing agent and the information used for price personalisation can influence perceived pricing fairness. Consumers may perceive AI pricing agents as fairer than human pricing agents because AI systems are viewed as less self-interested, less judgmental, and more consistent in their application of pricing rules (Duani et al., 2024; Zhang and Cheng, 2024). At the same time, behavioural information may be perceived as a more acceptable basis for personalisation than demographic information because it is more closely related to consumers' own behaviour and is more controllable. In contrast, demographic characteristics such as age, gender or location are generally less controllable and may be perceived as less relevant for price differentiation (Priester et al., 2020; Heidary et al., 2024; Sun and Tang, 2021).

Attribution theory provides a basis for expecting these factors to interact. Consumers evaluate not only the outcome of a pricing decision but also infer the motives underlying the pricing process (Campbell, 1999). These inferences may depend on both the agent responsible for the decision and the information used to determine the price. Thus, the perceived fairness of an AI pricing agent may depend on whether the underlying basis for personalisation is itself viewed as acceptable.

When behavioural information is used, consumers may perceive the basis of the personalised price as more relevant and legitimate because it reflects previous consumer behaviour. In this context, the perceived consistency and reduced personal judgement associated with AI may reinforce the perception that the pricing procedure is fair. When demographic information is used, however, concerns about the controllability and legitimacy of the information may become more salient. These concerns may weaken the extent to which the perceived objectivity of AI translates into a fairness advantage over human pricing agents.

This reasoning suggests that the AI fairness advantage should not be equally strong across different personalisation bases. Rather, the advantage should be stronger when prices are based on behavioural information than when they are based on demographic information.

H3: Personalization basis moderates the effect of pricing agents on perceived fairness. Specifically, the AI fairness advantage is stronger when prices are based on behavioral information and weaker when prices are based on demographic information.

4. Experiment

To test the research hypotheses, a 2×2 between-subjects vignette experiment was conducted. The design was based on the experimental design of Duani et al. (2024), specifically referring to Study 1, extended by the 2×2 structure used in Zhang and Cheng (2024). The survey was administered in English via Qualtrics using convenience sampling and the data were analysed using a 2×2 between-subjects ANOVA.

4.1 Design & Participants

The design included two factors, each with two levels, resulting in four conditions in total. Every participant saw only one version of the survey. Furthermore, each participant saw only one version of the survey and was randomly assigned to that condition.

Condition	Pricing Agent	Personalization Basis
1	AI Algorithm Pricing Agent	Behavioral (previous travel experiences and past travel choices)
2	AI Algorithm Pricing Agent	Demographic (gender, age and nationality)
3	Human Pricing Agent	Behavioral (previous travel experiences and past travel choices)
4	Human Pricing Agent	Demographic (gender, age and nationality)

Table 1. Summary of the conditions.

The travel-insurance scenario was inspired by Duani et al. (2024), particularly Study 1, which compared algorithmic and human pricing agents in an insurance-pricing context and examined perceived price fairness. An insurance context was selected because it represents a relatively high-stakes purchase, making fairness concerns particularly relevant. The present study adapts this design to travel insurance and introduces a second experimental factor: whether the personalised price is based on behavioural or demographic consumer information. All participants received a fixed personalised quote of €120.

The use of a 2×2 design combining pricing agent and personalisation basis was shaped by the empirical research design of Zhang and Cheng (2024), although their second factor concerned advantaged versus disadvantaged pricing rather than consumer data type. The measurement of perceived pricing fairness was based on the three-item measure used by Zhang and Cheng (2024), while the two-item purchase intention measures were based on Duani et al. (2024). The wording of both measures was adapted to the present travel-insurance scenario. The fairness questions referred to the personalised price and its determination, and the purchase intention questions referred to the intention of participants to purchase the insurance at the presented price.

Furthermore, I conducted a priori power analysis in G*Power (Faul et al., 2007) to determine the required sample size for this vignette study. Using a medium effect size of $f = 0.25$ (Cohen, 1977), 80% power, and $\alpha = 0.05$, the corresponding analysis output indicated a minimum of 128 participants. This converts to approximately 32 participants per condition. To account for incomplete responses or a possible uneven distribution across the four conditions, 152 participants were targeted, resulting in an evenly spread distribution of 38 participants per condition.

Participants were recruited using a convenience sampling approach. I distributed a single Qualtrics survey link, which randomly assigned respondents to one of the four experimental conditions. The link has been distributed through several online channels, including LinkedIn, Instagram, WhatsApp and the researcher's personal network. The survey was available between 19/07/2026 and 31/07/2026, during which reminder messages were distributed through the same recruitment channels to encourage participation. Participation was voluntary and no financial compensation was offered. Despite these recruitment efforts, the final sample remained below the target established by the a priori power analysis. The questionnaire and all experimental materials were administered in English.

4.2 Procedure

Participants first received information about the purpose of the study and were informed that participation was voluntary and anonymous. Before proceeding, participants accepted the consent terms and confirmed that they met the eligibility requirements, including being at least 18 years of age and sufficiently proficient in English. Participants were then randomly assigned to one of the four experimental conditions. They subsequently read a vignette describing the purchase of a travel insurance policy, in which they received a personalised insurance quote of €120. Depending on the assigned condition, participants were informed that the price had been determined either by an AI pricing system or a human pricing analyst using either behavioural or demographic information.

Exact Vignette and Manipulation Wording

All participants first read an identical scenario describing a hypothetical travel insurance purchase, reproduced in full below, before receiving the condition-specific manipulation sentence that varied the pricing agent and personalization basis.

Imagine you recently booked a jungle hike group expedition trip to the Tayrona National Park in Colombia. Based on the duration of the trip (3 weeks) and considering that it is your first time visiting South America, you decide to purchase a full coverage travel insurance policy. You contact an insurance provider to receive a personalized quote. During the insurance application process, you are asked to provide information about yourself and the specific trip by answering a couple of questions in an online form. The mandatory survey includes questions about whether accidents occurred during previous trips, past travel choices, and requests information about your gender, age, and nationality. The next day, you receive a call from the insurance provider, which tells you that the (total package) insurance quote is €120.

Depending on the randomly assigned condition, participants then read one of the following four sentences, which independently manipulated the pricing agent (AI vs. human) and the personalization basis (behavioral vs. demographic). The exact wording for each condition is reported in Table 2.

Condition	Pricing Agent	Personalization Basis	Exact Manipulation Wording (presented after the common scenario text)
1	AI	Behavioral	"The employee of the insurance provider informs you that the established quote is determined by an AI system of the company using information about your previous travel experiences and past travel choices to set up the insurance price."
2	AI	Demographic	"The employee of the insurance provider informs you that the established quote is determined by an AI system of the company using information about your gender, age, and nationality to set up the insurance price."
3	Human	Behavioral	"The employee of the insurance provider informs you that the established quote is determined by a human pricing analyst of the company using information about your previous travel experiences and past travel choices to set up the insurance price."
4	Human	Demographic	"The employee of the insurance provider informs you that the established quote is determined by a human pricing analyst of the company using information about your gender, age, and nationality to set up the insurance price."

Table 2. *Exact manipulation wording by experimental condition. All four conditions received an identical scenario description before the condition-specific sentence shown here.*

After reading the scenario, participants completed the measures of perceived pricing fairness and purchase intention, followed by the manipulation checks. Finally, participants answered several exploratory questions regarding their experience with AI chatbots and personalised pricing and were given the opportunity to provide an optional open-ended explanation of their fairness evaluation. Afterwards, participants were thanked for their participation.

4.3 Measures

The survey consisted of legitimate measurement scales, adapted from prior research. Unless otherwise indicated, all items were measured using seven-point Likert scales ranging from 1 (*strongly disagree*) to 7 (*strongly agree*).

Perceived pricing fairness

Perceived pricing fairness served as the primary dependent variable and was measured using the three-item scale developed by Zhang and Cheng (2024). The questions tested whether participants perceived the personalised insurance pricing procedure as fair, acceptable, and appropriate. Responses to the three items were averaged to create a fairness score, with higher scores indicating greater perceived pricing fairness.

Purchase intention

Purchase intention served as the secondary dependent variable and was measured using a two-item scale adapted from Duani et al. (2024). Both questions tested the willingness and likelihood of participants to purchase the presented travel insurance policy. The average of the two items was used to construct the purchase intention measure, with higher scores reflecting stronger purchase intentions.

Manipulation checks

Two manipulation checks were included to assess whether participants correctly identified the experimental manipulations. Participants indicated whether the insurance price had been determined by an AI pricing system or a human pricing analyst and whether the price was based on behavioural or demographic information. In addition, participants assessed the perceived realism of the vignette study using a seven-point scale ranging from 1 (*very unrealistic*) to 7 (*very realistic*).

Exploratory measures

Finally, participants reported how frequently they interacted with AI chatbots and how often they believed they had previously encountered personalised pricing. An optional open-ended question invited participants to explain the reasoning behind their fairness evaluation of the personalised price. These measures were included for exploratory purposes and to provide additional context for interpreting the quantitative findings.

4.4 Results

Data Screening, Scale Reliability and Manipulation Checks

Prior to hypothesis testing, the data were screened for incomplete responses. Of the 95 responses recorded, 65 were fully completed and formed the primary analytic sample (Human/Behavioral $n = 15$, Human/Demographic $n = 15$, AI/Behavioral $n = 18$, AI/Demographic $n = 17$). Scores for perceived pricing fairness and purchase intention were calculated by averaging the corresponding questionnaire items, after confirming that all items were positively worded and required no reverse-scoring. The fairness scale showed strong internal consistency (3 items, Cronbach's $\alpha = .90$), as did the purchase intention scale (2 items, $\alpha = .91$), both reliability estimates calculated from the present sample rather than adopted from prior studies.

Manipulation checks indicated that 52 of 65 participants (80.0%) correctly identified the pricing agent and 47 of 65 (72.3%) correctly identified the personalization basis used in their condition; 41 (63.1%) passed both checks. These results are presented by condition in Table 3. Because a substantial proportion of participants failed one or both checks, the primary analyses reported below include all complete and eligible responses regardless of manipulation-check performance. Supplementary robustness analyses restricted to participants who correctly answered the check relevant to each specific hypothesis are reported later in Table 5.

Pricing Agent	Data Type	n	Agent check correct (%)	Data-type check correct (%)	Both correct (%)
Human	Behavioral	15	60.0	73.3	53.3
Human	Demographic	15	73.3	60.0	53.3
AI	Behavioral	18	94.4	94.4	88.9
AI	Demographic	17	88.2	58.8	52.9

Table 3. Percentage of participants who correctly answered each manipulation check, by condition, in the primary analytic sample ($N = 65$). Blue shading indicates AI conditions; orange shading indicates Human conditions.

Descriptive Statistics

Table 4 presents perceived fairness and purchase intention across the four experimental conditions. Descriptively, participants in the AI–behavioral condition reported the highest fairness ratings ($M = 4.33$, $SD = 1.63$), while participants in the human–demographic condition reported the lowest ($M = 3.36$, $SD = 0.95$). The remaining two conditions fell between these extremes.

Condition	n	Fairness M (SD)	Purchase Intention M (SD)
AI / Behavioral	18	4.33 (1.63)	4.39 (1.97)
AI / Demographic	17	4.04 (1.32)	3.88 (1.21)
Human / Behavioral	15	4.07 (1.57)	4.23 (1.59)
Human / Demographic	15	3.36 (0.95)	3.00 (1.31)

Table 4. Perceived fairness and purchase intention by condition. Results are based on the 7-point Likert scale. Blue shading indicates AI conditions; orange shading indicates Human conditions.

Testing Perceived Pricing Fairness

A 2 (Pricing Agent: AI vs. Human) $\times$ 2 (Personalization Basis: Behavioral vs. Demographic) between-subjects ANOVA was conducted on perceived fairness.

The estimated mean fairness rating was higher in the AI conditions ($M = 4.19$, 95% CI [3.71, 4.66]) than in the human conditions ($M = 3.71$, 95% CI [3.20, 4.22]), although this main effect was not statistically significant, $F(1, 61) = 1.85$, $p = .179$, partial $\eta^2 = .03$. As Hypothesis 1 predicted that AI-generated pricing would be perceived as fairer than human-generated pricing, H1 was not supported in the primary analysis.

The estimated mean fairness rating was higher in the behavioural conditions ($M = 4.20$, 95% CI [3.71, 4.69]) than in the demographic conditions ($M = 3.70$, 95% CI [3.20, 4.19]), although this main effect was not statistically significant, $F(1, 61) = 2.07$, $p = .155$, partial $\eta^2 = .03$. As the primary analysis did not provide statistically significant evidence that behavioural information resulted in higher perceived pricing fairness than demographic information, H2 was not supported.

Lastly, the pricing agent $\times$ personalization-basis interaction was not statistically significant, $F(1, 61) = 0.36$, $p = .553$, partial $\eta^2 = .01$. H3, which predicted that the AI fairness advantage would be stronger when prices were based on behavioural rather than demographic information, was therefore not supported. Because the interaction itself was not statistically significant, the simple effects are not interpreted as evidence of moderation.

Supplementary Robustness Analysis

A supplementary robustness analysis was conducted among the 52 participants who correctly identified the pricing agent. Within this restricted sample, the estimated fairness ratings differed significantly between the AI and human pricing-agent conditions in the predicted direction, $F(1, 48) = 4.27$, $p = .044$, partial $\eta^2 = .08$ (see Table 5). This result provides additional evidence regarding the predicted AI–human difference among participants who correctly identified the pricing agent. However, because the analysis was based on a restricted subsample defined by performance on the manipulation check, it does not replace the primary analysis or establish that the nonsignificant primary result was caused by manipulation-check failure.

Hypothesis / Outcome	Robustness sample	n	F	df	p	partial η^2
H1 – Agent (Fairness)	Agent-check passers	52	4.27	1, 48	.044	.08
H1 – Agent (Purchase Intention)	Agent-check passers	52	1.54	1, 48	.220	.03
H2 – Data Type (Fairness)	Data-check passers	47	3.14	1, 43	.083	.07
H2 – Data Type (Purchase Intention)	Data-check passers	47	4.22	1, 43	.046	.09
H3 – Interaction (Fairness)	Both-checks passers	41	0.39	1, 37	.538	.01
H3 – Interaction (Purchase Intention)	Both-checks passers	41	0.32	1, 37	.572	.01

Table 5. *Supplementary robustness analyses using subsamples defined by manipulation-check performance. H1 analyses include participants who correctly identified the pricing agent, H2 analyses include participants who correctly identified the personalisation basis, and H3 analyses include participants who correctly identified both manipulations. The primary analyses were conducted on the full sample (N = 65) and are reported separately.*

Note. Bold values indicate statistically significant effects at $p < .05$.

Secondary Outcome

For the secondary outcome, purchase intention, the estimated mean was higher in the AI conditions ($M = 4.14$, 95% CI [3.61, 4.66]) than in the human conditions ($M = 3.62$, 95% CI [3.05, 4.19]), although this main effect was not statistically significant, $F(1, 61) = 1.78$, $p = .187$, partial $\eta^2 = .03$. Personalisation basis, however, produced a statistically significant main effect: participants reported higher purchase intentions when pricing was based on behavioural information ($M = 4.31$, 95% CI [3.76, 4.86]) than when it was based on demographic information ($M = 3.44$, 95% CI [2.89, 3.99]), $F(1, 61) = 5.00$, $p = .029$, partial $\eta^2 = .08$. The interaction between pricing agent and personalisation basis on purchase intention was not statistically significant, $F(1, 61) = 0.87$, $p = .354$, partial $\eta^2 = .01$. These findings are considered secondary because purchase intention was not the primary outcome used to test H1–H3.

Figures showing the estimated marginal means and 95% confidence intervals for fairness and purchase intention across all four conditions are presented below.

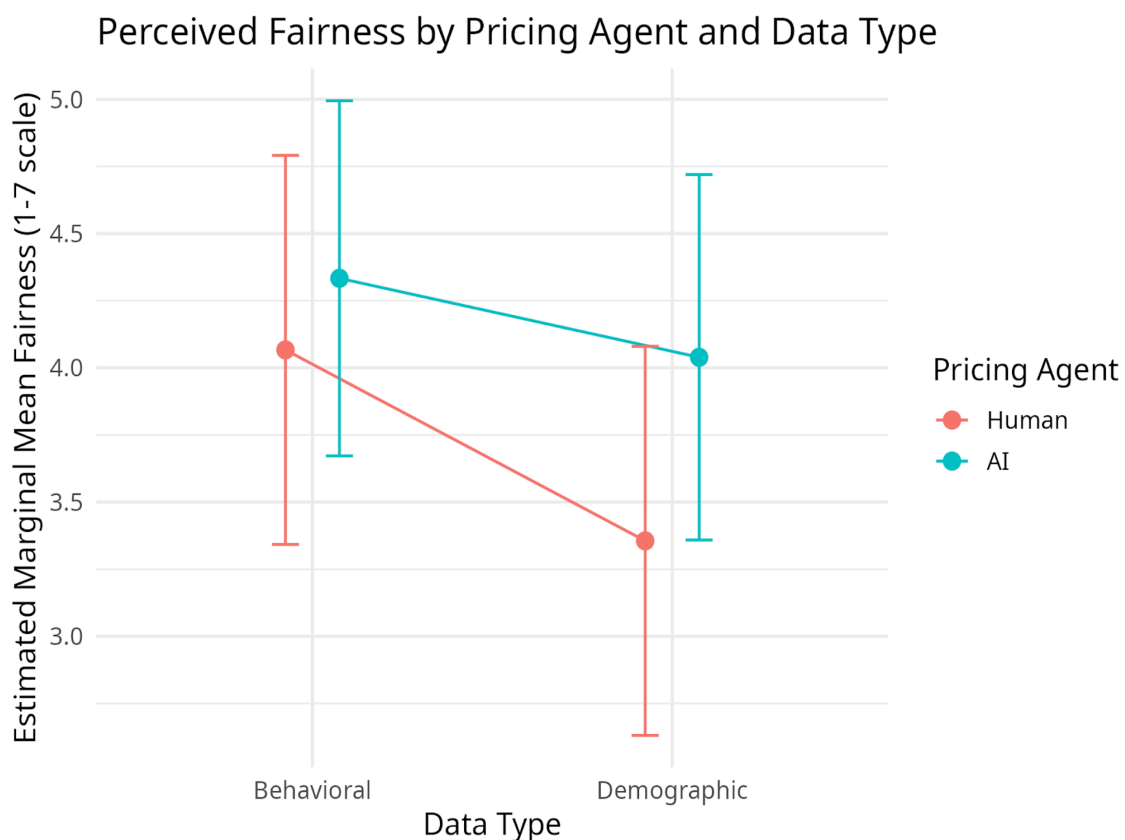


Figure 1: *Estimated marginal means and 95% confidence intervals for fairness across all four conditions.*

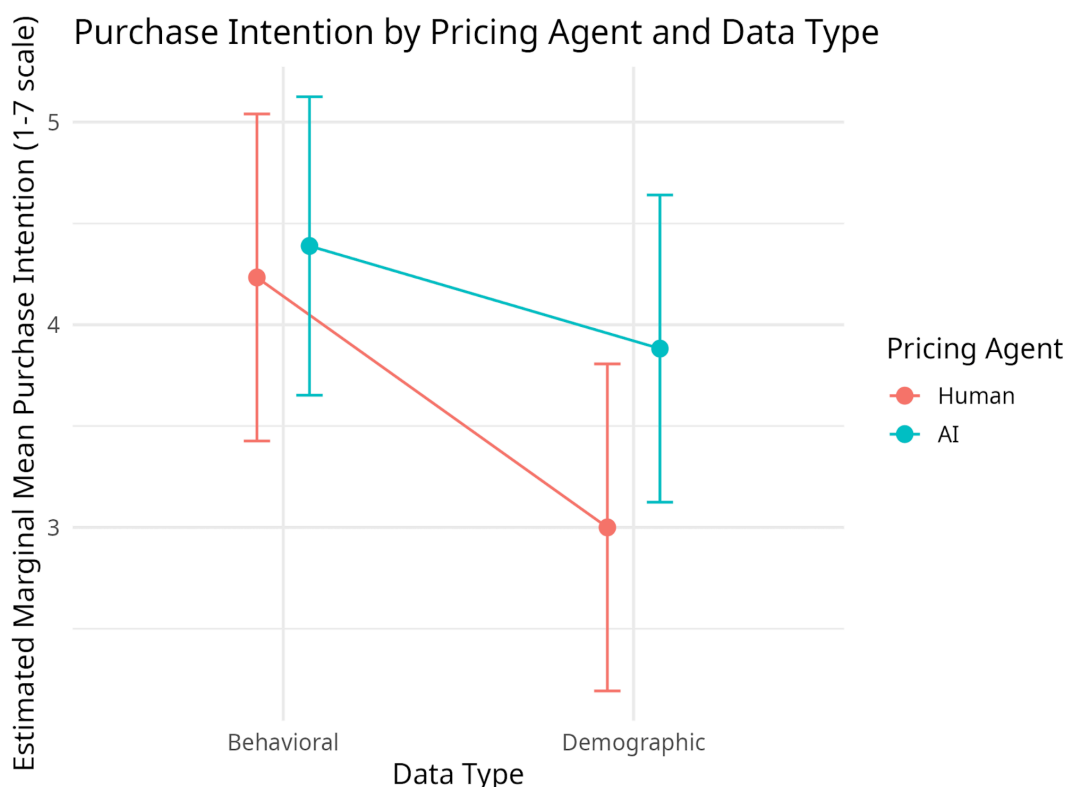


Figure 2: *Estimated marginal means and 95% confidence intervals for purchase intention across all four conditions.*

4.5 Discussion

The purpose of this experiment was to examine the extent to which the AI fairness advantage in personalised pricing depends on the type of information used to personalise the price. In the primary analysis, none of the three hypothesised effects on perceived pricing fairness reached statistical significance. The direction of both main effects was consistent with the theoretical predictions: AI-determined prices and behaviourally personalised prices received higher estimated fairness ratings than their respective comparison conditions. However, these differences were not statistically significant, and the predicted interaction between pricing agent and personalisation basis also showed no evidence of the hypothesised moderation pattern.

Hypothesis 1 was not supported in the primary analysis. The main effect of the pricing agent on perceived fairness did not reach statistical significance, $F(1, 61) = 1.85$, $p = .179$, partial $\eta^2 = .03$. The estimates provide suggestive but inconclusive evidence in the predicted

direction, with AI-generated pricing having a higher estimated mean fairness rating than human-generated pricing. A supplementary robustness analysis restricted to the 52 participants who correctly identified the pricing agent yielded a significant effect in the predicted direction, $F(1, 48) = 4.27$, $p = .044$, partial $\eta^2 = .08$. This result should be interpreted as supplementary evidence rather than as evidence that the primary non-significant result was caused by manipulation-check failure, as the restricted analysis was based on a different subsample.

The estimated difference between the AI and human conditions in the present study was 0.48 points on the seven-point fairness scale. This difference is smaller than the AI–human difference reported in Study 1 of Duani et al. (2024), where price fairness was higher in the algorithm condition ($M = 3.29$) than in the human condition ($M = 2.55$), a difference of 0.74 points. The AI-human difference was therefore larger in their Study 1 than in the present primary analysis. Duani et al. (2024) used an insurance-pricing scenario with a more explicit algorithm-versus-human manipulation, whereas the present study used a travel-insurance vignette and additionally manipulated the type of consumer information used for personalisation. Their study also distinguished between price fairness and process fairness, whereas the present study focused on perceived pricing fairness as the primary outcome. These differences in context, manipulation, and measurement mean that the differences in effect size should be interpreted cautiously.

The findings are also consistent with Zhang and Cheng (2024), who found that the perceived fairness of AI versus human pricing depended on the pricing context. Their results showed that disadvantaged prices were perceived as fairer when determined by an algorithm, whereas this pattern was reversed for advantaged prices. This suggests that an AI fairness advantage may depend not only on the pricing agent but also on the nature of the resulting price. The present study did not explicitly manipulate advantaged versus disadvantaged pricing, which may partly explain why the AI vs human difference was not statistically significant in the primary analysis.

Hypothesis 2 was also not supported for perceived pricing fairness. The main effect of personalisation basis did not reach statistical significance, $F(1, 61) = 2.07$, $p = .155$, partial $\eta^2 = .03$. The estimates provide suggestive but inconclusive evidence in the predicted direction, with behavioural information associated with a higher estimated mean fairness rating than demographic information. This direction is consistent with previous research suggesting that consumers may evaluate behavioural information more favourably than demographic information when used for personalised pricing (Heidary et al., 2024; Priester et al., 2020;

Sun and Tang, 2021). However, the present study does not provide statistically significant evidence that personalisation basis affected perceived pricing fairness.

For the secondary outcome of purchase intention, personalisation basis did produce a statistically significant main effect, $F(1, 61) = 5.00$, $p = .029$, partial $\eta^2 = .08$. Participants reported higher purchase intentions when prices were based on behavioural rather than demographic information. As purchase intention was not the primary outcome used to test H1–H3, this result should be interpreted as a secondary finding rather than as support for H2. Nevertheless, it suggests that the type of consumer information used in personalised pricing may influence consumer responses beyond perceived pricing fairness.

Hypothesis 3 was not supported. The interaction between pricing agent and personalisation basis on perceived pricing fairness was not statistically significant, $F(1, 61) = 0.36$, $p = .553$, partial $\eta^2 = .01$. Therefore, the present study does not provide evidence that the AI–human difference in perceived pricing fairness depends on whether behavioural or demographic information is used for price personalisation. The hypothesised moderation effect was not observed in the study sample.

These findings should be interpreted in light of the study's limited statistical power. The achieved sample of 65 participants was substantially below the approximately 128 participants required to detect a conventional small-to-medium effect ($f = .25$), with 80% power under the original design assumptions. The limited sample size reduced the ability of the empirical study to detect relatively small effects. At the same time, the absence of statistically significant effects in the primary analyses should not be interpreted as evidence that the hypothesised relationships do not exist. The supplementary robustness result for H1 should likewise not be interpreted as establishing that the primary non-significant result was caused by failing manipulation checks. Rather, the difference between the primary and restricted analyses indicates that the relationship deserves further investigation with larger samples and more robust designs.

Overall, the primary analyses did not provide statistically significant support for the hypothesised effects of pricing agent, personalisation basis, or their shared interaction on perceived pricing fairness. The estimates for both main effects were in the predicted direction, while the significant secondary effect of personalisation basis on purchase intention suggests that consumer responses to personalised pricing may extend beyond fairness perceptions. The findings therefore provide inconclusive evidence for the proposed relationships and highlight the need for further research using larger samples and different pricing contexts.

5. General Discussion

5.1 Main Findings

The present study examined whether the perceived fairness advantage of AI pricing over human pricing depends on the type of consumer data used to personalise prices. Contrary to expectations, the primary analyses did not provide statistically significant support for the hypothesised effects of pricing agent, personalization basis or their interaction on perceived pricing fairness. Although AI-generated personalised prices and prices based on behavioural information were evaluated more favourably on average than human-generated prices and prices based on demographic information, these differences were not statistically significant.

A robustness analysis restricted to participants who correctly identified the pricing agent indicated that AI-generated pricing was perceived as significantly fairer than pricing determined by a human pricing analyst. Furthermore, while behavioural personalization did not significantly increase perceived pricing fairness, it resulted in a higher purchase intention than demographic personalization. No statistically significant moderation effect was observed between pricing agent and personalization basis. Consequently, the findings of this study do not provide sufficient empirical evidence that the relationship between pricing agent and perceived pricing fairness depends on whether personalised prices are based on behavioural or demographic consumer data.

Therefore, the main research question can be answered as follows: within the travel insurance scenario examined in this study, no statistically significant evidence was found that the AI fairness advantage depends on the type of consumer data used for personalised pricing.

5.2 Contributions of Findings

First, this study provides a nuanced perspective on previous findings regarding the AI fairness advantage in personalised pricing. Contrary to earlier research (Duani et al., 2024), the empirical primary analyses did not reveal a statistically significant difference in perceived pricing fairness considering AI-generated pricing and pricing determined by a human pricing analyst. However, the robustness analysis restricted to participants who correctly identified the pricing agent did provide evidence of an AI fairness advantage. This supplementary finding indicates that the observed AI vs human difference may warrant further investigation under conditions in which participants clearly recognise the pricing agent.

Secondly, the present study contributes to the literature on consumer data used for personalised pricing by showing that behavioural personalization resulted in significantly higher purchase intention than demographic personalization, despite no statistically significant differences in perceived pricing fairness. Previous research has suggested that demographic information, such as age, location or ethnicity, is generally perceived as a less acceptable basis for personalised pricing than behavioural information, such as purchase history or previous consumer behaviour (Priester et al., 2020; Heidary and Pluut, 2025; Heidary et al., 2024). The present findings suggest that the effects of consumer data type may differ across outcome measures, with the observed difference emerging for purchase intention but not for perceived fairness.

Lastly, this study contributes to the personalised pricing literature by integrating two research streams that have largely been examined separately. Previous research has primarily investigated either the influence of the pricing agent (AI versus human) on perceived pricing fairness or the role of consumer data type in personalised pricing (Duani et al., 2024; Zhang and Cheng, 2024; Priester et al., 2020; Heidary and Pluut, 2025; Heidary et al., 2024; Sun and Tang, 2021). To the best of our knowledge, limited empirical research has examined the joint influence of both factors within a single experimental framework. Although the present study did not find statistically significant evidence that personalization basis moderates the relationship between pricing agent and perceived pricing fairness, it provides one of the first empirical examinations of this relationship. The absence of a significant interaction does not provide evidence for the hypothesised moderation effect, but the relationship may warrant further investigation in larger samples and different pricing contexts. Consequently, the present findings offer a foundation for future research examining the combined effects of pricing agent and personalization basis on consumer evaluations of personalised pricing.

5.3 Practical Implications

The findings of this study have practical implications for firms that apply personalised pricing based on consumer information rather than dynamic pricing mechanisms. Examples include insurance providers, banks offering personalised financial products (including mortgages) and vehicle leasing companies. However, because none of the primary fairness effects were statistically significant, the results should be interpreted as informing managerial considerations rather than providing direct recommendations for a particular pricing approach.

Organisations should carefully consider both the pricing agent responsible for determining personalised prices and the type of consumer information used during the pricing process. Although the present study did not find statistically significant differences in perceived pricing fairness between behavioural and demographic personalization, behavioural personalization resulted in significantly higher purchase intention. Together with previous research, this suggests that firms should carefully evaluate which types of consumer information are used when implementing personalised pricing, as behavioural information is generally perceived as a more acceptable basis for pricing than demographic information (Priester et al., 2020; Heidary and Pluut, 2025; Heidary et al., 2024). However, these findings do not establish a general fairness advantage for behavioural information. The results also cannot be generalized to all personalized pricing settings.

If firms continue to rely on demographic information for personalised pricing, providing consumers with clear explanations regarding why particular information is collected and how it contributes to the pricing decision may improve perceptions of procedural fairness. Increasing transparency surrounding the pricing procedure may therefore help to reduce concerns regarding unfair treatment or price discrimination. However, the present study did not directly test the effect of transparency, so this implication should be considered for future research and practice rather than as an established effect.

The findings also provide limited implications for the use of AI in personalised pricing. The primary analysis did not provide statistically significant evidence that AI-generated prices were perceived as fairer than human-generated prices. The supplementary robustness analysis among participants who correctly identified the pricing agent did show an effect in the predicted direction. This is in line with previous research (Duani et al., 2024; Zhang and Cheng, 2024) but does not serve as sufficient evidence for a general AI fairness advantage since the finding is based on a restricted subsample. Firms should therefore avoid assuming that communicating the use of AI will automatically increase perceptions of fairness. Instead, the fairness implications of AI-based pricing should be evaluated in every specific pricing context in which the system is used.

Overall, the findings suggest that managers should recognise that consumers evaluate not only the outcome of personalised pricing but also the procedure through which prices are established. Although the present study did not find statistically significant evidence that pricing agent and personalisation basis jointly influence perceived pricing fairness, the observed differences in purchase intention and the supplementary robustness findings indicate that these factors warrant further consideration. Organisations should therefore carefully consider how personalised pricing procedures are designed and communicated,

while recognising that additional research is needed before firm recommendations can be generalised across different industries and other pricing contexts.

5.4 Limitations

Several limitations of the present study should be acknowledged. First, the empirical study relied on a convenience sample rather than a random sampling technique. Participants were recruited through a survey link shared within the network of the researcher, meaning that the sample may not be representative of the broader consumer population. Consequently, the findings should be interpreted with caution, as they may not be generalisable to all consumers or personalised pricing contexts.

Furthermore, the questionnaire was administered exclusively in English. Although this enabled participation from respondents with different national backgrounds, it may have limited participation among individuals with lower English proficiency. Differences in language proficiency may also have influenced how participants interpreted the vignette and questionnaire items.

Another limitation concerns the experimental design. The study employed a hypothetical vignette in which participants imagined purchasing travel insurance rather than making an actual purchase decision. Although vignette experiments provide a high degree of experimental control and enable causal relationships to be examined, they cannot fully replicate real purchasing situations in which consumers face actual financial consequences. Participants may therefore respond differently in real market environments than in a hypothetical scenario.

Relatedly, the experiment measured perceived pricing fairness and purchase intention rather than actual consumer behaviour. Although purchase intention provides an indication of the behaviour of consumer responses, it does not necessarily translate into actual purchasing decisions. Real-world purchasing behaviour is influenced by numerous additional factors. Examples include competing offers, price differences between providers, budget constraints and individual preferences, which were beyond the scope of the present study.

In addition, the experiment focused exclusively on a travel insurance scenario. Although this context was considered appropriate for studying personalised pricing, the findings cannot automatically be generalised to other industries in which personalised pricing is applied, such as banking, vehicle leasing or airline services. Consumer evaluations of pricing fairness may differ across products and services depending on the purchasing context. Furthermore, the

findings are limited to personalised pricing settings and cannot automatically be generalised to other pricing strategies, such as dynamic pricing, because these rely on different pricing mechanisms and decision-making processes.

Finally, the achieved sample size was lower than initially planned. Recruitment relied on voluntary participation without financial incentives, making it difficult to reach the target number of respondents identified in the a priori power analysis. Consequently, the statistical power of the study was reduced, increasing the likelihood that smaller effects remained undetected. The lack of statistically significant effects in the primary analyses should therefore not necessarily be interpreted as evidence that such effects do not exist. Rather, the present study may not have been sufficiently powered to detect them.

5.5 Future Research

Future research should examine whether the findings of the present study can be replicated across different industries and product and service categories in which personalised pricing is applied. Considering the limited sample size and the absence of statistically significant effects in the primary analyses, future studies should also investigate whether these findings reflect context-specific boundary conditions or whether larger samples reveal the hypothesised relationships. Although the present study focused on travel insurance, consumer evaluations of pricing fairness may differ across other contexts, such as banking, vehicle leasing or airline services. Furthermore, future studies could employ larger-scale vignette experiments or field experiments in which participants make actual purchasing decisions rather than responding to hypothetical scenarios. Such research would increase the external validity of the findings while providing greater insight into whether perceived pricing fairness translates into actual purchasing.

Future research could also investigate different demographic and behavioural consumer data types used for personalised pricing. While the present study compared these data types at a general level, future studies could examine the effects of specific behavioural variables, such as purchase history, browsing activity, social media engagement and geolocation, as well as different demographic characteristics. This would also partly address the research agenda proposed by Duani et al. (2024), who noted that limited empirical evidence exists regarding the responses of consumers to the use of increasingly available consumer data for personalised pricing. Such research would provide a more detailed understanding of which types of consumer information are perceived as more or less acceptable when determining personalised prices.

Another promising direction for future research concerns the role of explanations and transparency surrounding algorithmic pricing. Future studies could investigate whether providing consumers with information about how AI pricing systems operate, why particular consumer information is used, and how pricing decisions are reached influences perceived pricing fairness. Comparing different forms of AI and human pricing justifications may also provide additional insights into how communication strategies influence consumer responses.

Additionally, future research should further examine the psychological mechanisms that influence consumers' fairness evaluations. Variables such as perceived discrimination, perceived exploitation and intentionality may help explain why consumers respond differently to AI and human pricing agents or to different consumer data types. Investigating these mechanisms would provide a deeper understanding of how consumers evaluate personalised pricing procedures and would further extend the application of attribution theory within the personalised pricing literature.

Finally, future studies could collect more detailed participant characteristics, including demographic information such as age, nationality and country of residence. This would enable researchers to conduct more robust cross-cultural and generational comparisons, providing greater insight into whether consumer perceptions of AI pricing and personalised pricing fairness differ across countries, cultures and demographic groups.

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7. Appendix

Complete scenario survey:

Base scenario:

Imagine you recently booked a jungle hike group expedition trip to the Tayrona National Park in Colombia. Based on the duration of the trip (3 weeks) and considering that it's your first time visiting South-America, you decide to purchase a full coverage travel insurance policy.

You contact an insurance provider to receive a personalized quote. During the insurance application process, you are asked to provide information about yourself and the specific trip by answering a couple of questions in an online form. The mandatory survey includes questions about whether accidents occurred during previous trips, past travel choices and requests information about your gender, age and nationality.

The next day, you receive a call of the insurance provider which tells you that the (total package) insurance quote is €120.

Condition-specific manipulation paragraph which was added after the base scenario:

Group 1 — AI + Behavioral: *“The employee of the insurance provider informs you that the established quote is determined by an AI system of the company using information about your previous travel experiences and past travel choices to set up the insurance price.”*

Group 2 — AI + Demographic: *“The employee of the insurance provider informs you that the established quote is determined by an AI system of the company using information about your gender, age and nationality to set up the insurance price.”*

Group 3 — Human + Behavioral: *“The employee of the insurance provider informs you that the established quote is determined by a human pricing analyst of the company using information about your previous travel experiences and past travel choices to set up the insurance price.”*

Group 4 — Human + Demographic: *“The employee of the insurance provider informs you that the established quote is determined by a human pricing analyst of the company using information about your gender, age and nationality to set up the insurance price.”*

Dependent variables tested, all using the 7-point scales (1 = strongly disagree, 7 = strongly agree):

- *"The pricing I received is fair"*
- *"The pricing I received is acceptable"*
- *"The pricing I received is appropriate"*

- *"I would be likely to purchase this travel insurance policy from this provider."*
- *"I would be willing to buy this travel insurance policy from this provider."*

Manipulation checks:

- *"Who determined the price you paid?"* AI algorithm / (Human) Pricing Analyst / don't know)

- *"What information was used to set your price?"* (Previous travel experiences and past travel choices / Gender, age, nationality / don't know)

- *"How realistic was this scenario?"*. With a scale between 1–7, (1 = very unrealistic / 7 = Very realistic)

Explanatory variables all using the 7-point scales, (1 = Not at all / 7 = Very often):

"How often do you interact with artificial chatbots (ChatGPT, Claude, Gemini)?"

"How often do you think that you paid a price which was personalized for you?"

The final question requested the participant for a written explanation to further elaborate on the previously answered choices regarding price fairness. The last screen informed participants that their response had been recorded and were thanked for contributing to the empirical research after the final question was completed or skipped.

Condition	Pricing Agent	Personalization Basis
1	AI Algorithm Pricing Agent	Behavioral (previous travel experiences and past travel choices)
2	AI Algorithm Pricing Agent	Demographic (gender, age and nationality)
3	Human Pricing Agent	Behavioral (previous travel experiences and past travel choices)
4	Human Pricing Agent	Demographic (gender, age and nationality)

Table 1: Summary of the conditions. Each participant was randomly assigned to one condition only and saw one version of the survey.

Condition	Pricing Agent	Personalization Basis	Exact Manipulation Wording (presented after the common scenario text)
1	AI	Behavioral	"The employee of the insurance provider informs you that the established quote is determined by an AI system of the company using information about your previous travel experiences and past travel choices to set up the insurance price."
2	AI	Demographic	"The employee of the insurance provider informs you that the established quote is determined by an AI system of the company using information about your gender, age, and nationality to set up the insurance price."
3	Human	Behavioral	"The employee of the insurance provider informs you that the established quote is determined by a human pricing analyst of the company using information about your previous travel experiences and past travel choices to set up the insurance price."
4	Human	Demographic	"The employee of the insurance provider informs you that the established quote is determined by a human pricing analyst of the company using information about

			your gender, age, and nationality to set up the insurance price.”
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Table 2. Exact manipulation wording by experimental condition. All four conditions received an identical scenario description before the condition-specific sentence shown here.

Pricing Agent	Data Type	n	Agent check correct (%)	Data-type check correct (%)	Both correct (%)
Human	Behavioral	15	60.0	73.3	53.3
Human	Demographic	15	73.3	60.0	53.3
AI	Behavioral	18	94.4	94.4	88.9
AI	Demographic	17	88.2	58.8	52.9

Table 3. Percentage of participants who correctly answered each manipulation check, by condition, in the primary analytic sample (N = 65). Blue shading indicates AI conditions; orange shading indicates Human conditions.

Condition	n	Fairness M (SD)	Purchase Intention M (SD)
AI / Behavioral	18	4.33 (1.63)	4.39 (1.97)
AI / Demographic	17	4.04 (1.32)	3.88 (1.21)
Human / Behavioral	15	4.07 (1.57)	4.23 (1.59)
Human / Demographic	15	3.36 (0.95)	3.00 (1.31)

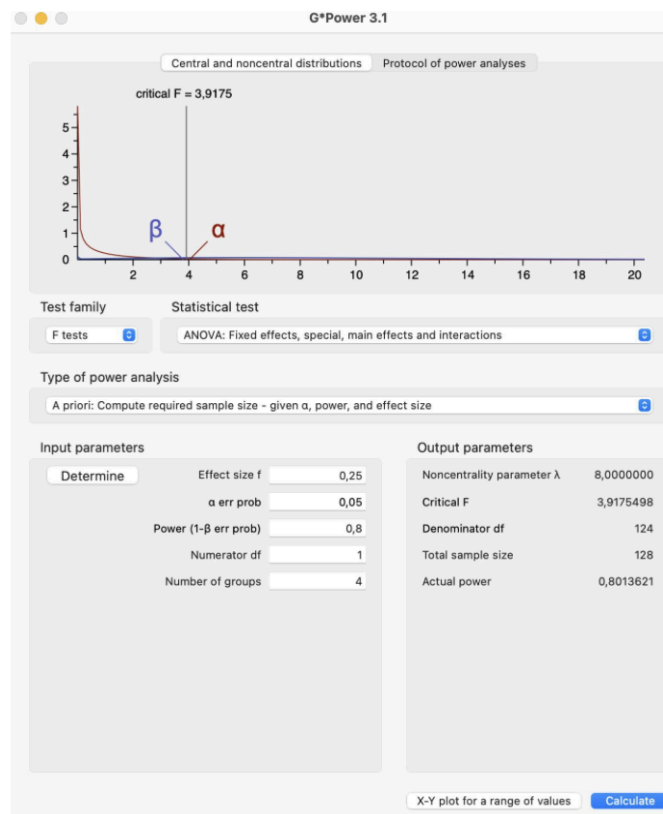
Table 4. Perceived fairness and purchase intention by condition. Results are based on the 7-point Likert scale. Blue shading indicates AI conditions; orange shading indicates Human conditions.

Hypothesis / Outcome	Robustness sample	n	F	df	p	partial η^2
H1 – Agent (Fairness)	Agent-check passers	52	4.27	1, 48	.044	.08

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H2 – Data Type (Purchase Intention)	Data-check passers	47	4.22	1, 43	.046	.09
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Table 5. Supplementary robustness analyses using subsamples defined by manipulation-check performance. H1 analyses include participants who correctly identified the pricing agent, H2 analyses include participants who correctly identified the personalisation basis, and H3 analyses include participants who correctly identified both manipulations. The primary analyses were conducted on the full sample ($N = 65$) and are reported separately.

Note. Bold values indicate statistically significant effects at $p < .05$.



G*Power results: Additional screenshot

Additional statistical output

The following figures reproduce the estimated marginal means and 95% confidence intervals reported in the Results section. Figure 1 presents perceived pricing fairness across the four experimental conditions, while Figure 2 presents purchase intention across the four conditions.

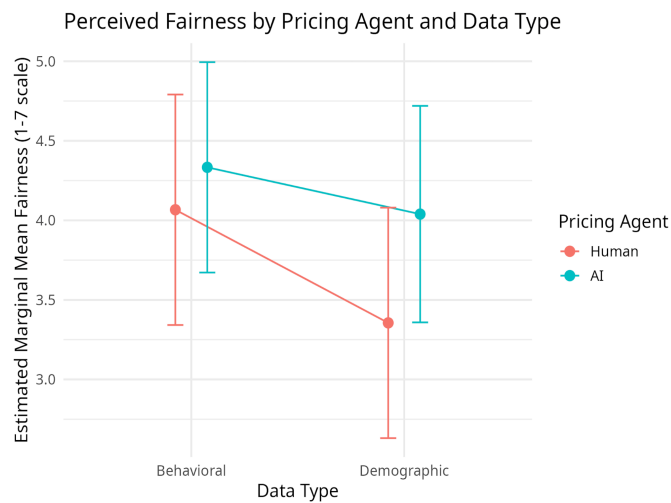


Figure 1: *Estimated marginal means and 95% confidence intervals for fairness across all four conditions.*

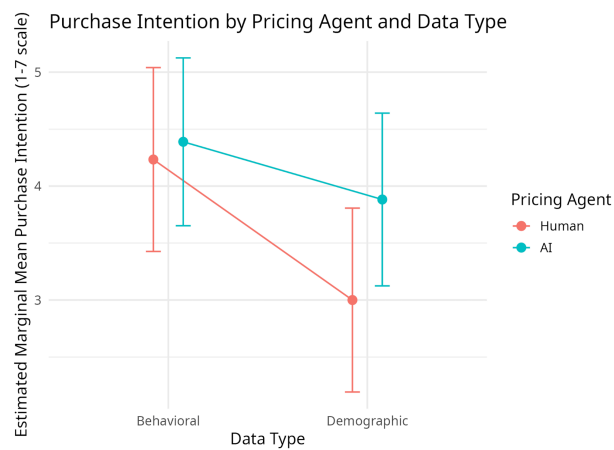


Figure 2: *Estimated marginal means and 95% confidence intervals for purchase intention across all four conditions.*